

Do Syntactic Attention Heads Matter for Prediction?

A Multilingual Dependency and Masking Study of mBERT and GPT-2

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Abstract. This study asks whether interpretable attention heads in transformers are only useful for describing model behavior, or whether they also actively *support prediction*. We combine two analyses. First, mBERT attention heads are evaluated as dependency-graph predictors across **20 Universal Dependencies treebanks** and **16,244 filtered sentences**. Second, ranked head subsets are retained under attention masking in both mBERT and GPT-2 medium. Grammar-sensitive mBERT heads recur mainly in middle and upper-middle layers (L5–L8), but the same head does not always encode the same relation across languages. Prediction efficiency depends strongly on model objective: in mBERT, syntactic-relation heads at $k=96$ yield only **1.28% NLL overhead**, while GPT-2 medium requires positional or hybrid heads. Ablation with layer-matched random controls confirms that targeted categories are genuinely more necessary than chance. Interpretability, sufficiency, and causal importance must be assessed independently.

1 Motivation

Transformer models are frequently described as having “learned syntax.” This claim actually conflates two separate questions: (1) *interpretability* — whether some attention heads align with linguistic structure like dependency arcs; and (2) *prediction utility* — whether those heads actively help the model predict words.

A head can look grammatical without being prediction-critical, and a head can be prediction-critical without mapping to any clean linguistic relation. Prior work in BERTology [2, 1] has largely focused on (1). Our study directly addresses (2) by retaining or removing linguistically-ranked head subsets and measuring the impact on prediction quality.

Core research question: Do dependency-sensitive attention heads carry usable grammatical information for prediction, and under what conditions are such heads *sufficient* and *efficient* for maintaining language-model quality?

2 Hypotheses

We separate four ideas that are easy to conflate:

1. **Recoverability.** Does a head’s argmax attention edge recover dependency arcs beyond a positional baseline?
2. **Sufficiency.** Can a retained subset alone preserve near-full-model prediction?
3. **Efficiency.** How many heads are needed to reach a target NLL overhead threshold?
4. **Necessity.** Does removing a targeted category hurt more than removing a layer-matched random subset?

H1 (Cross-language localization). Grammar-sensitive heads should concentrate in middle and upper-middle layers and recur across typologically different languages.

H2 (Prediction sufficiency). Syntactic or structural head subsets should preserve prediction within 10% and 5% NLL overhead thresholds.

H3 (Prediction efficiency). The most efficient category should differ by model objective: syntactic-relation heads for mBERT;

Study design: from interpretable heads to prediction-efficient masked submodels

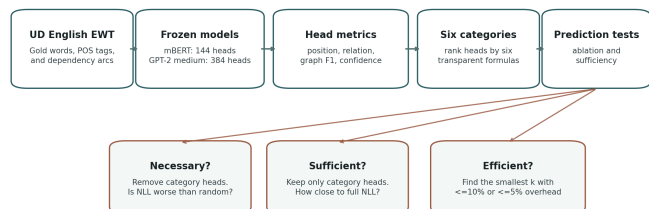


Figure 1: Study pipeline from UD dependency data through head characterization, six ranking categories, and prediction experiments (sufficiency, ablation, efficiency).

positional or hybrid for GPT-2.

H4 (Necessity beyond random). Ablating targeted categories should damage NLL more than ablating layer-matched random subsets of equal size.

3 Methods

3.1 Data and Models

The multilingual dependency analysis covers **20 UD treebanks** and **16,244 filtered sentences** spanning English, Finnish, Indonesian, Hindi, Urdu, Arabic, Chinese, Dutch, French, German, Italian, Japanese, Korean, Latin, Romanian, Sanskrit, Spanish, Tamil, Turkish, and Vietnamese. The masking study uses English UD-EWT so both models are compared on the same annotation scheme.

mBERT (bert-base-multilingual-cased): 12 layers $\times$ 12 heads = 144 total attention heads. Bidirectional attention; tokens can attend left and right.

GPT-2 medium: 24 layers $\times$ 16 heads = 384 total attention heads. Causal masking; tokens can only attend to earlier positions.

Run sizes are verified in Table 1.

Table 1: Experiment verification.

Component	Evidence
mBERT sufficiency	169 rows; 6,209 tokens; 1,357 sentences
mBERT ablation	49 rows; 6,209 tokens; 1,357 sentences
GPT-2 sufficiency	313 rows; 1,406 sentences; 18,562 words
GPT-2 ablation	73 rows; 1,406 sentences; 18,562 words
Random controls	3 seeds per condition; layer distribution preserved

Six retained-head rankings used in the masking tests

Positional $s_h = \text{norm}(p_h)$ Ranks heads whose top attention repeatedly follows the same relative token offset.	Syntactic relation $s_h = .60\text{norm}(M_h) + .40\text{norm}(n_h)$ M_h : summed positive relation margins. n_h : number of supported relation directions.
Pure syntactic $s_h = s_h^{\text{pos}}$, only if $p_h < .80$ Keeps syntactic heads but removes heads that are mainly near-position shortcuts.	Structural graph $s_h = \text{norm}(F1_h^{\text{graph}})$ Scores unlabeled dependency-edge recovery, ignoring UD relation labels.
Syntax-position hybrid $s_h = .50s_h^{\text{pos}} + .50s_h^{\text{pos}}$ Tests whether prediction prefers a mix of grammar signal and local-offset behavior.	Syntax composite $s_h = .45s_h^{\text{pos}} + .35s_h^{\text{pos}} + .20\text{norm}(a_h)$ A broad score combining graph recovery, relation margins, and attention confidence.

Figure 2: Six head-ranking formulas. These are scoring rules, not mutually exclusive linguistic classes; the same head may rank highly in multiple categories.

3.2 Dependency Graph Extraction

For each layer ℓ and head h , we extract the attention matrix $A^{(\ell,h)}$, where $A_{ij}^{(\ell,h)}$ is the softmax-normalized attention weight from token i to token j . A head’s predicted dependency parent for word i is:

$$\hat{p}_h(i) = \arg \max_{j \neq i} A_{ij}^{(\ell,h)}. \quad (1)$$

WordPiece subwords are mapped back to words before comparing with UD arcs. Both arc directions are evaluated: dependent-to-head (*dep-to-head*) and head-to-dependent (*head-to-dep*).

3.3 Positional Baseline and Syntactic Margin

Many dependency relations are locally predictable. A head always attending to the previous token can appear grammatical without encoding syntax. To control for this, we compare each head against the best fixed-offset baseline for the same relation and direction:

$$\Delta(h, r, d) = \text{Acc}(h, r, d) - \max_o \text{Acc}(\text{offset}=o, r, d). \quad (2)$$

A head-relation pair is labelled *syntactic* only when $\Delta(h, r, d) \geq 0.10$. This is the primary guard against mistaking word-order habits for grammar.

3.4 Six Head Ranking Categories

Figure 2 shows the six categories used for head masking:

1. **Positional** — heads that attend to the same relative offset consistently.
2. **Syntactic relation** — heads scoring high on summed relation margins and relation count.
3. **Pure syntactic** — syntactic heads with positional share < 0.80 (genuine grammar, not position shortcuts).

4. **Structural graph** — heads with high unlabeled dependency-edge F1, ignoring UD relation labels.
5. **Syntax-position hybrid** — 50/50 blend of syntactic and positional scores.
6. **Syntax composite** — 0.45 structural + 0.35 syntactic + 0.20 confidence.

3.5 Masking, Ablation, and Random Controls

For sufficiency, the top- k heads of a category are kept active; all others are masked. For ablation, those k heads are removed from the full model. NLL overhead versus the full model is:

$$\text{overhead}(C) = 100 \left(\frac{\text{NLL}(C)}{\text{NLL}_{\text{full}}} - 1 \right). \quad (3)$$

Ablation effects are compared against *layer-matched random controls* (3 seeds per condition) so that results are not confounded by some layers being globally more important:

$$\epsilon_{\text{vs. rand}} = \Delta\text{NLL}_{\text{targeted}} - \overline{\Delta\text{NLL}}_{\text{random}}. \quad (4)$$

Positive ϵ means the targeted category is more necessary than a comparably distributed random subset.

4 Results

4.1 Cross-Language Dependency Heads

The cross-language analysis covers 20 treebanks (Figure 3, Table 2). The main finding is not a single universal parser head, but a reusable mid-layer grammar region whose specific relation targets shift by language.

Table 2: mBERT syntactic heads by language. A full 20-language bar chart is in Figure 3.

Language	Sent.	Heads	Best margin / relation
Latin	1,400	118	0.500 (advmod:lmod)
German	731	114	0.826 (compound:prt)
Sanskrit	1,400	96	0.316 (advcl:cond)
Dutch	672	94	0.718 (compound:prt)
Italian	444	80	0.483 (conj)
Korean	752	80	0.350 (conj)
Chinese	363	80	0.602 (cop)
English	1,392	77	0.429 (advcl:relcl)
Finnish	1,197	72	0.500 (nsubj:outer)
Turkish	813	53	0.313 (nsubj)
Indonesian	1,000	50	0.500 (csubj:pass)
Hindi	1,386	35	0.542 (ccomp)
Arabic	421	33	0.302 (mark)
Urdu	1,000	14	0.581 (ccomp)
Japanese	364	7	0.159 (advcl)

Layer concentration. Extracted head coordinates cluster in layers L5–L8 (Figure 4), confirming H1. Grammar-sensitive attention is not randomly distributed; it is a mid-network property.

Language-family signatures. German and Dutch show the largest single-relation margins (+0.826 and +0.718 for `compound:prt`), reflecting the strong separable-particle and auxiliary patterns of Germanic languages. German L7–H3 achieves 97.1% accuracy on this relation against a 14.5% positional baseline. Chinese encodes copula and object strongly

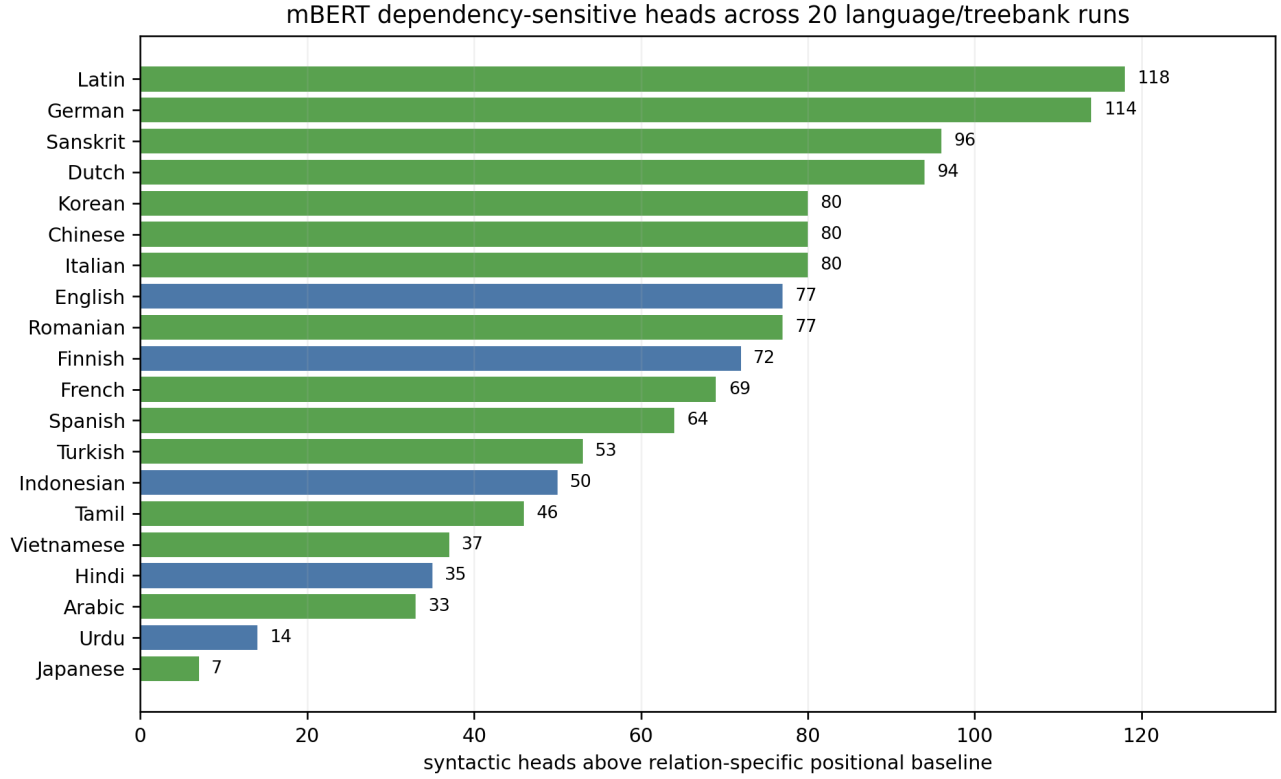


Figure 3: mBERT syntactic heads across 20 language/treebank runs (16,244 sentences total). A head is counted once per relation-direction that beats the relation-specific positional baseline by +0.10 or more. Latin, German, and Sanskrit lead; Japanese and Urdu are the sparsest.

despite having less inflectional morphology than European languages, with L5–H8 reaching a +0.602 margin on *cop*. Japanese is the sparsest outlier (7 syntactic heads), with modest margins around *advcl* and *mark*.

Reusable heads. Within the five-language masking subset (English, Finnish, Hindi, Indonesian, Urdu), L5–H6, L8–H5, L3–H0, L8–H6, and L4–H5 appear syntactic across all five languages. Crucially, the *coordinate* is stable but the *relation* is flexible: L8–H5 recovers conjunction in Italian, Korean, and French, but shifts to adverbial clauses in Japanese.

4.2 Prediction Efficiency Under Head Masking

The masking experiments answer a different question from the dependency analysis: *given only a subset of heads, how well can the model still predict?*

mBERT headline result. Syntactic-relation heads at $k=96$ (66.7% of 144 heads) achieve NLL 3.883, only **1.28% overhead** versus the full model (NLL 3.834), retaining 98.4% of top-1 accuracy. This simultaneously satisfies the 5% and 10% NLL thresholds. No other category reaches the 5% threshold at this budget.

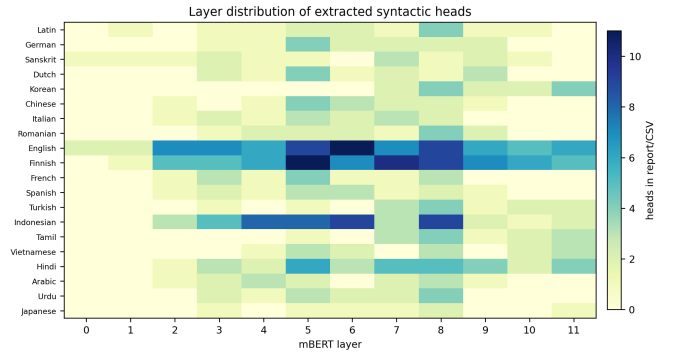


Figure 4: Layer distribution of extracted syntactic heads across all 20 languages. Darker cells indicate more heads extracted from that layer. Concentration in L5–L8 is clear across typologically diverse languages.

GPT-2 headline result. Positional heads reach the 10% threshold earliest ($k=128$, 33.3% of 384 heads, 9.04% overhead). At larger budgets, syntax-position hybrid is the best quality choice: $k=256$ gives 3.50% overhead; $k=288$ gives **1.96% overhead**. Pure syntactic and syntactic-relation categories never reach 10% overhead even at $k=288$ (stalling at 11.13%).

The contrast between mBERT (syntactic-first) and GPT-2 (positional/hybrid-first) directly reflects the difference in training objectives. Full threshold decisions are in Table 3.

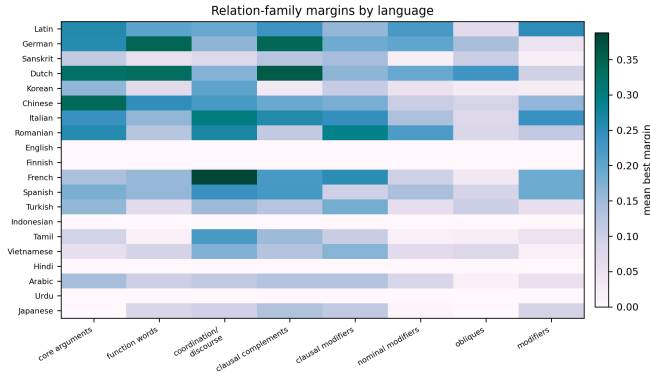
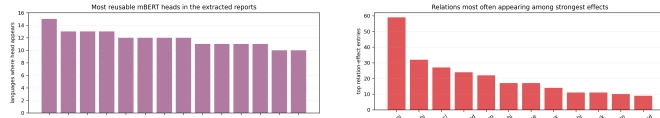


Figure 5: Relation-family margins by language. Darker cells indicate where the best heads beat positional baselines most strongly. Germanic languages dominate the coordination and function-word columns.



(a) Most reusable mBERT heads.

(b) Most recurring relation effects.

Figure 6: Left: head coordinates that appear as syntactic across the most languages. Right: UD relation families most often showing positive margins in the extracted evidence.

4.3 Accuracy Recovery

Top-1 accuracy recovery (Figure 9) mirrors the NLL findings. For mBERT, syntactic-relation at $k=96$ retains **98.4%** of full-model accuracy. GPT-2 pure syntactic categories stall at 77.2% retention even at the largest budget, confirming that relation-aligned heads alone cannot carry causal next-token prediction. In contrast, GPT-2 requires positionally-aware heads to recover strong accuracy: at $k=288$, the syntax-position hybrid category reaches **94.7%** retention, while positional heads alone reach similar levels slightly earlier. This further demonstrates that for causal language models, sequential positional information is paramount for recovering next-token prediction performance.

4.4 GPT-2 Causal Ceiling

GPT-2 cannot attend to future tokens, so many UD arcs are structurally inaccessible. On the English UD-EWT evaluation set, only **39.6%** of gold edges are directly recoverable by left-to-right attention (Figure 10, left). Relations where the head appears to the right of the dependent (*obj*, *obl*) have high ceilings; relations where the head typically precedes (*det*, *aux*, *nsubj*) have near-zero ceilings. GPT-2’s low absolute syntax F1 is partly architectural, not a learning failure.

4.5 Ablation and Random Controls

Ablation results (Figure 11) support H4. For GPT-2 at $k=32$: syntax composite $\epsilon = +1.33$; hybrid $\epsilon = +0.68$; positional $\epsilon = +0.79$. For mBERT at $k=16$: structural graph $\epsilon = +0.48$; positional $\epsilon = +0.40$.

The key distinction is between necessity and sufficiency:

Table 3: Prediction-efficiency threshold decisions.

Model	Category	k	%act.	NLL	Overhead
mBERT	Syntactic relation	96	66.7	3.883	1.28%
mBERT	Syntax composite	108	75.0	4.032	5.17%
mBERT	Structural graph	108	75.0	4.175	8.89%
mBERT	Positional (best)	72	50.0	5.813	51.6%
GPT-2	Positional ($\leq 10\%$)	128	33.3	5.104	9.04%
GPT-2	Hybrid ($\leq 5\%$)	256	66.7	4.844	3.50%
GPT-2	Hybrid (best)	288	75.0	4.772	1.96%
GPT-2	Pure syntactic (best)	288	75.0	5.202	11.1%

mBERT positional heads: strongly necessary (ablating 16 hurts +0.40 beyond random) but catastrophically insufficient alone (51.6% NLL overhead when kept alone). A clear case of necessary-but-not-sufficient.

mBERT syntactic-relation heads: highly sufficient collectively (1.28% overhead at $k=96$) but less individually necessary at small k ($\epsilon \approx 0$ at $k=4$). Distributed redundancy: the circuit exists and carries prediction, but is spread across enough heads that no single head is uniquely load-bearing.

5 Theoretical Implications

5.1 Attention Encodes Real, Distributed Syntax

The multilingual evidence rules out the null hypothesis that interpretable heads are merely positional regularities relabelled as grammar. Margins like +0.826 on German *compound:prt* (97.1% accuracy vs. 14.5% baseline) are large, consistent effects replicated across typologically diverse scripts and grammatical frameworks. The evidence covers SOV and SVO word orders, agglutinative and fusional morphology, and scripts ranging from Latin to Devanagari to Hangul to Chinese characters.

However, this syntactic structure is distributed rather than localized. No single head acts as a universal parser. Instead, a reusable mid-layer region (L5–L8) hosts heads that are recruited for grammatical tasks, with the specific relation recovered varying by language. L8–H5 encodes conjunction in Italian, Korean, and French but shifts to adverbial clauses in Japanese. This flexibility argues for *reusable computation routines* rather than hard-coded syntactic slots. The model develops general-purpose grammar-sensitive attention mechanisms during pre-training that are applied to whichever dependency structures are statistically prevalent in each language.

5.2 Interpretability and Predictive Utility Are Dissociated

This project empirically separates two properties that are often conflated. A head that aligns with a dependency relation is *interpretable*. Whether it actually carries prediction-relevant information is a separate question requiring sufficiency and necessity testing.

In mBERT, the two properties happen to co-occur: syntactic-relation heads are both interpretable and prediction-efficient. Retaining 66.7% of heads ranked by grammatical sensitivity preserves 98.4% of masked-token accuracy. This is a strong result: the subset defined by linguistic criteria is also close to sufficient for the model’s pretraining objective.

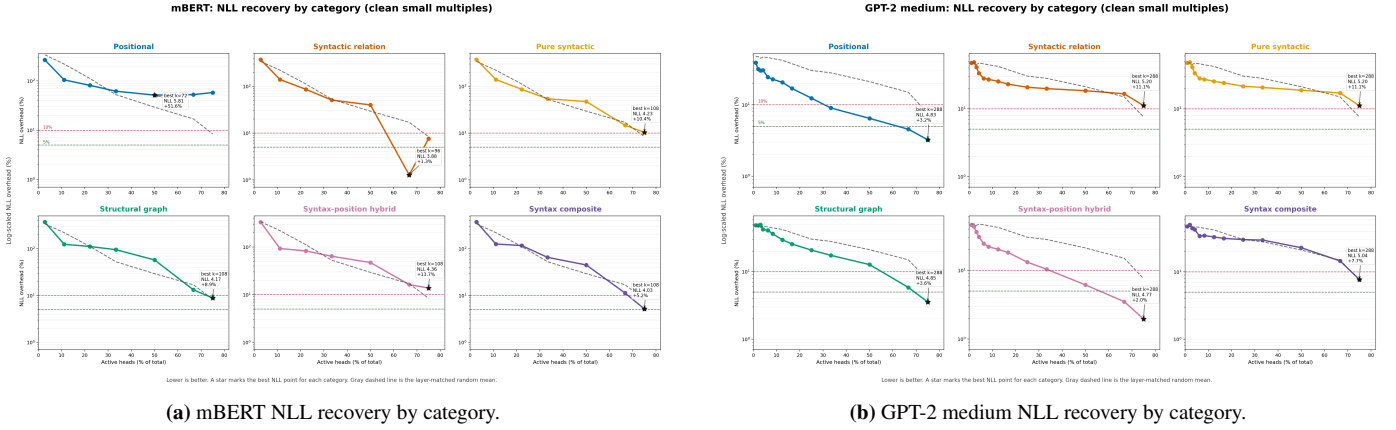


Figure 7: NLL overhead vs. active head fraction for each ranking category (log-scaled y-axis). A star marks the best NLL point per category; the gray dashed line marks the layer-matched random mean. mBERT is best preserved by syntactic-relation heads; GPT-2 medium requires positional or syntax-position hybrid heads.

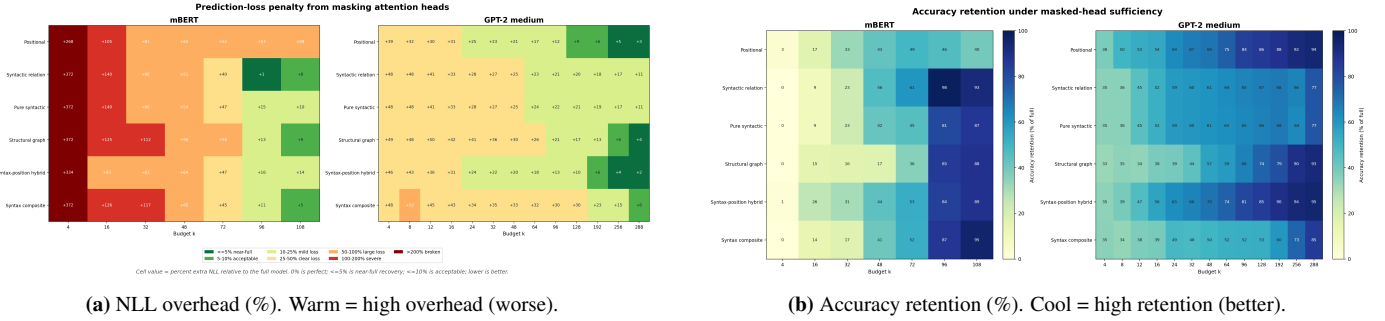


Figure 8: Performance heatmaps for each category and budget k across mBERT (left block) and GPT-2 medium (right block). mBERT syntactic-relation stands out clearly in the cool (low-overhead) region at $k=96$.

In GPT-2, they decouple. Purely syntactic subsets cannot recover next-token prediction even at 75% retention, while positional and hybrid heads do so comfortably. Yet GPT-2 does encode grammar (the dependency analysis confirms this, within the causal ceiling). The point is that the *training objective determines which interpretable patterns become prediction-load-bearing*. A causal left-to-right objective rewards local sequential continuity more than long-range structural dependencies. Syntactic structure is present but not a primary bottleneck for next-token generation.

This has a direct methodological implication: finding that a head aligns with a linguistic structure does not imply it matters for the model’s objective. The converse is equally true. Both halves of this claim are supported in the data.

5.3 Predictive Sufficiency and the Training Objective

The mBERT sufficiency result is strong enough to be worth interpreting carefully. A subset defined purely by how well heads recover linguistic relations — without any reference to prediction quality — nearly sustains full masked-language-model performance. This suggests that the grammatically-ranked portion of mBERT’s attention mechanism is not a decorative overlay on top of prediction-oriented circuitry. It is, to a large degree, the prediction-oriented circuitry.

Why should this hold for masked-token prediction but not causal next-token prediction? When a model predicts a randomly masked word from bilateral context, the surrounding dependency structure strongly constrains the answer: what syntactic slot is this position filling, and what kind of word fills that slot? Syntactic heads that track those dependencies therefore carry prediction-relevant signal. A causal model predicting the next token faces a different problem: it primarily needs to know what follows the current context, which is dominated by local sequential patterns. Positional and hybrid heads that capture those patterns are therefore more load-bearing in GPT-2.

This interpretation is consistent with prior probing work (Tenney et al. 2019, Hewitt & Manning 2019) showing that BERT-like models develop richer syntactic representations than unidirectional models. Our contribution is to connect that interpretability result to *predictive utility* through sufficiency under masking. The richer syntactic representations are not just visible to probing classifiers; they are active participants in the forward computation.

5.4 Cross-Lingual Reuse and Representational Sharing

The cross-lingual reusability of head coordinates speaks to whether multilingual models develop genuinely language-neutral representations or simply language-specific routines

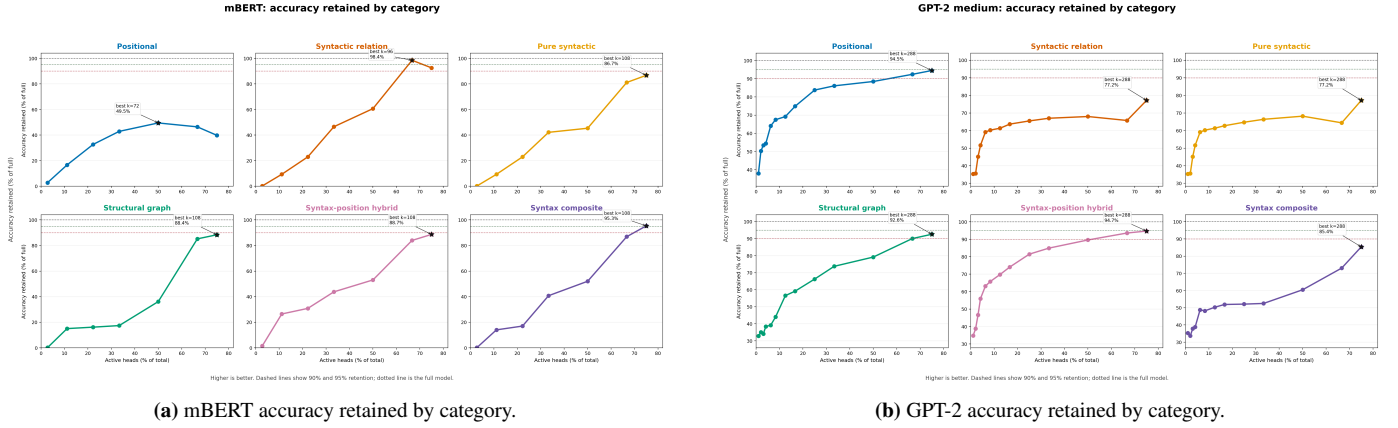


Figure 9: Accuracy retained as a percentage of the full model versus active head fraction. Dashed lines show 90% and 95% retention levels. The pattern mirrors NLL: mBERT syntactic-relation reaches 98.4% retention at $k=96$; GPT-2 hybrid reaches 94.7% at $k=288$.

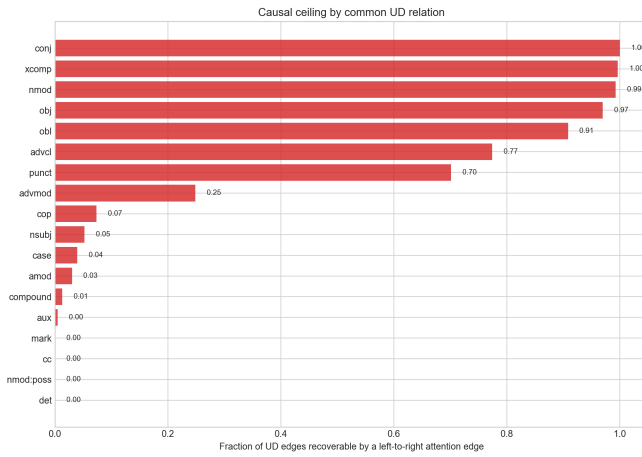


Figure 10: Left: fraction of gold UD edges directly recoverable by left-to-right attention, by relation. Right: syntax F1 vs. next-token accuracy across all head subsets. Syntax-efficient and prediction-efficient subsets occupy different regions.

that share parameters. Our evidence points to a middle position. The *coordinates* of syntactic heads are shared across languages (L5–H6, L8–H5, L3–H0, and L8–H6 are syntactic in all five project-run languages). But the *relations* recovered are flexible across languages.

This is consistent with mBERT developing general-purpose grammar-sensitive attention machinery that is applied differently in each language’s statistical environment. The concentration of reusable heads in layers L5–L8, rather than early processing layers (L0–L4) or final representation layers (L9–L11), suggests that cross-lingual structural generalization is a mid-network property. Fine-tuning and probing experiments targeting these layers may find the strongest cross-lingual transfer.

5.5 Practical Consequences for Model Compression

The sufficiency finding has a direct application. If 66.7% of mBERT’s attention heads — selected by syntactic-relation ranking — nearly sustains full prediction quality, then the remaining 33.3% are candidates for structured pruning. The key insight is

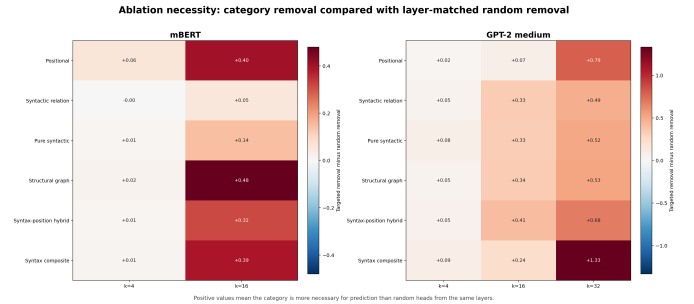


Figure 11: Targeted-minus-random ablation effects ($\epsilon_{\text{vs. rand}}$). Positive values confirm the category is more prediction-necessary than random heads from the same layers. GPT-2 effects grow sharply at $k=32$.

that the ranking criterion is determined by linguistic analysis, not by prediction loss on a held-out set. This provides a theoretically motivated compression strategy that is independent of the downstream task.

For GPT-2 and causal models more broadly, the analogous strategy would target positional and hybrid heads rather than purely syntactic ones. The practical implication is that optimal pruning strategies are objective-dependent: bidirectional models should be pruned syntactically; causal models should be pruned positionally/hybrid-first. This is a testable and potentially actionable finding for practitioners designing efficient language models.

6 Conclusion

This project delivers empirical answers to three interconnected questions about transformer attention heads:

- Do attention heads encode real grammar?** Yes, across 20 typologically diverse languages, with effects well above relation-specific positional baselines and concentrated in layers L5–L8.
- Are syntactic heads prediction-sufficient?** Yes for mBERT (1.28% NLL overhead at $k=96$); conditionally for GPT-2

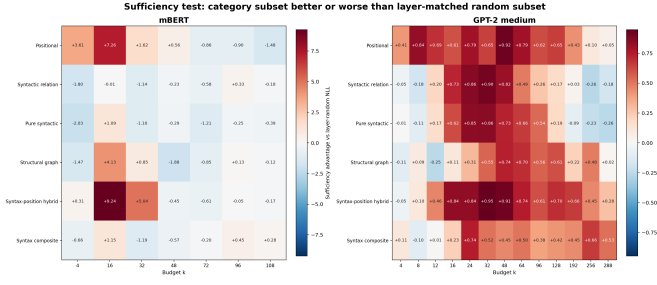


Figure 12: Sufficiency advantage over layer-matched random retained-head subsets. Positive (warm) cells mean the named category retains NLL better than random heads from the same layers.

(requires positional/hybrid heads; pure syntactic categories cannot carry causal prediction).

3. Are interpretability and prediction necessity the same?

No. They must be tested separately. mBERT shows alignment; GPT-2 shows dissociation.

The deeper theoretical point is that the training objective shapes which attention patterns become prediction-load-bearing. Bidirectional masked prediction makes syntactic representations useful for prediction. Causal next-token prediction shifts the load toward local positional and hybrid patterns. Interpretability is a starting point, not an endpoint. Sufficiency and necessity testing are required to determine whether an interpretable head actually matters for the model’s computation.

References

- [1] E. Voita, D. Talbot, F. Moiseev, R. Sennrich, and I. Titov, “Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned,” in *Proc. ACL*, 2019.
- [2] K. Clark, U. Khandelwal, O. Levy, and C. D. Manning, “What does BERT look at? An analysis of BERT’s attention,” in *Proc. BlackboxNLP*, 2019.
- [3] I. Tenney, D. Das, and E. Pavlick, “BERT rediscovers the classical NLP pipeline,” in *Proc. ACL*, 2019.
- [4] J. Hewitt and C. D. Manning, “A structural probe for finding syntax in word representations,” in *Proc. NAACL-HLT*, 2019.
- [5] S. Jain and B. C. Wallace, “Attention is not explanation,” in *Proc. NAACL-HLT*, 2019.
- [6] J. Nivre et al., “Universal Dependencies v2: An evergrowing multilingual treebank collection,” in *Proc. LREC*, 2020.
- [7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers,” in *Proc. NAACL-HLT*, 2019.
- [8] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language models are unsupervised multitask learners,” OpenAI Blog, 2019.
- [9] P. Michel, O. Levy, and G. Neubiger, “Are sixteen heads really better than one?” in *Proc. NeurIPS*, 2019.

Appendix A Additional Exploratory Figures

The following figures were obtained during the project but were not included in the main text.

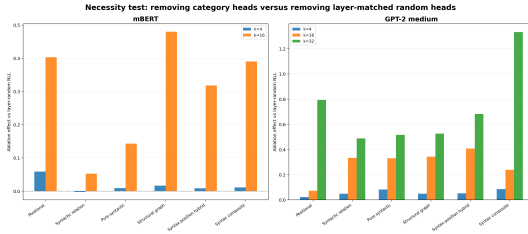


Figure 13: Ablation Effects Vs Random: Demonstrates the comparative impact of ablating targeted syntactic heads versus random baseline subsets.

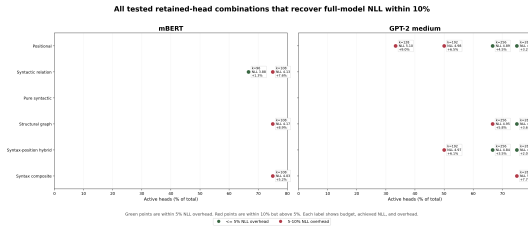


Figure 14: All Successful NLL Combinations: Highlights combinations of layers and heads that successfully maintain low Negative Log-Likelihood (NLL).

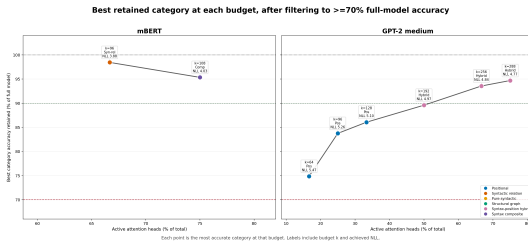


Figure 15: Best By Budget Accuracy 70Pct: Shows the optimal head configurations that maintain high accuracy for varying retention budgets.

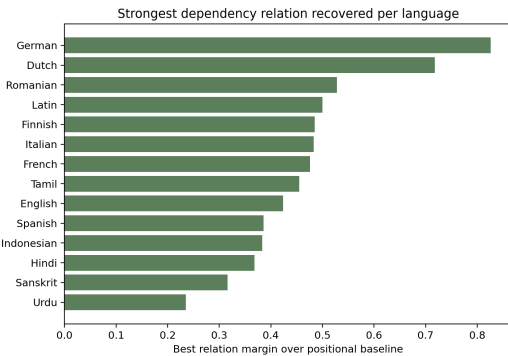


Figure 16: Best Relation Margin By Language: Compares the top relation prediction margins across different languages.

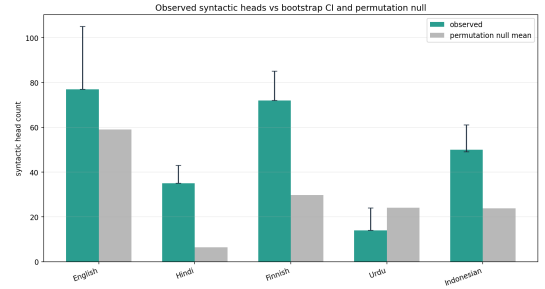


Figure 17: Bootstrap Permutation Syntactic Heads: Statistical significance testing using bootstrap permutations for syntactic head identification.

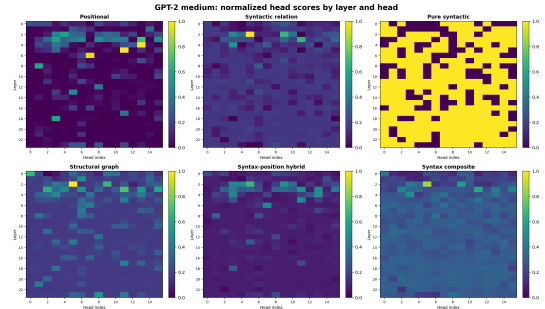


Figure 18: Gpt2 Head Score Maps: Spatial mapping of attention head scores showing the distribution of task-specific relevance.

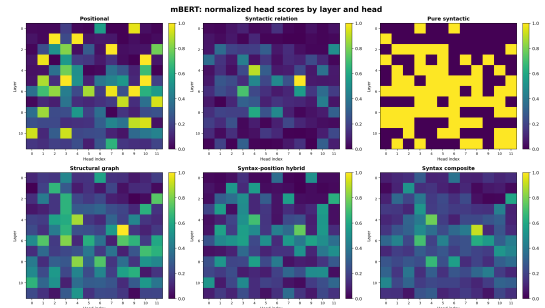


Figure 19: Mbert Head Score Maps: Spatial mapping of attention head scores showing the distribution of task-specific relevance.

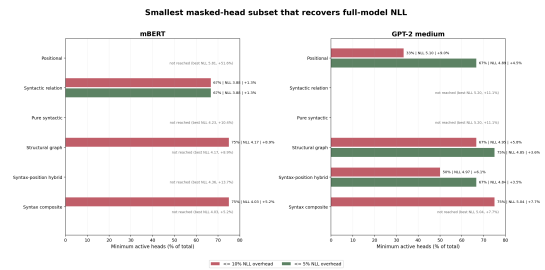


Figure 20: Minimum Budget Thresholds: Identifies the minimum number of heads required to achieve predefined performance thresholds.

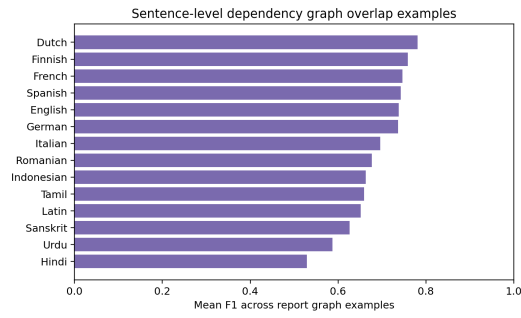


Figure 21: Example Graph F1 By Language: Sample dependency graph F1 performance across different languages.

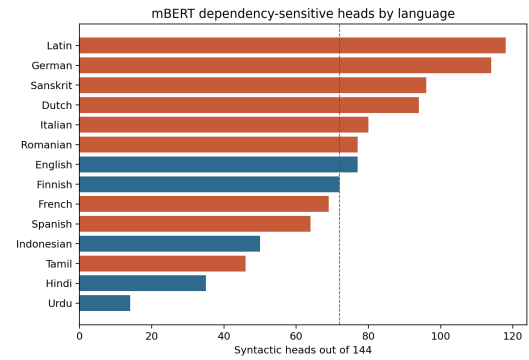


Figure 25: Syntactic Heads By Language: Distribution and counts of identified syntactic attention heads broken down by source language.

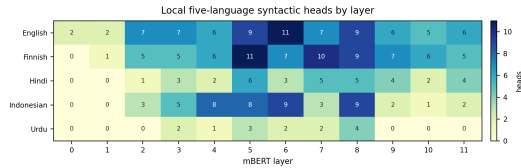


Figure 22: Local Mbert Syntactic Heads By Layer: Layer-wise breakdown of syntactically relevant heads specifically derived from local context.

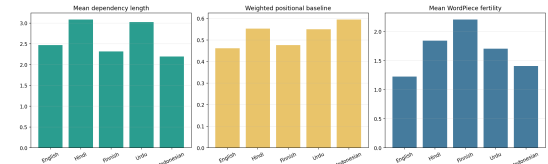


Figure 26: Typology Controls: Analysis controlling for typological differences across language families.

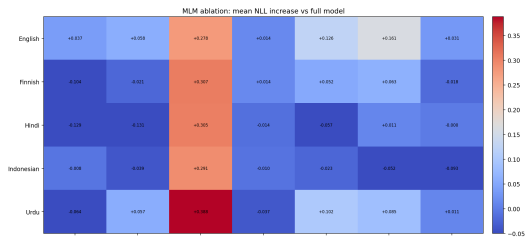


Figure 23: Mlm Ablation Delta Nll: Shows the delta Negative Log-Likelihood specifically for the Masked Language Modeling objective upon ablation.

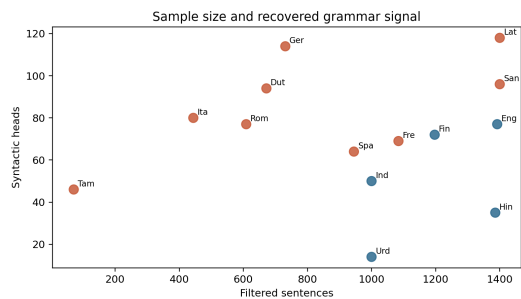


Figure 24: Sample Size Vs Syntax: Analyzes the tradeoff between dataset sample size and the accuracy of syntax recovery.